Element B: Documentation and Analysis of Prior Design Attempts

Two major design pathways were considered:

1. A two-joint arm with a rotating claw mechanism (inspired by OpenROV appendage modules)
2. A 6-thruster vertical/horizontal vector-based design with fixed tools

The first was abandoned due to fabrication time limits and high error rate in mechanical response underwater. Our team discovered that flexible joints with waterproof actuation posed significant failure points and severely slowed operation.

We also reviewed the BlueROV2 and determined its weight and cost were unsuitable for our budget and pool requirements. Our early tests with pool noodles and surface testing falsely suggested proper buoyancy, but deeper tests showed a net downward force due to tether drag (estimated at 1.4 N unaccounted). This instability flipped the robot during critical missions.

Documentation and images of these failures were used to guide the new build's simplification, focusing on propulsion and buoyancy rather than complex mechanical arms.